Yuyuan Han

Computational imaging & physics-aware learning

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PhD student in computational imaging and physics-aware learning from optical measurements.

Education & Research Appointments

Xi'an Jiaotong University · PhD student — 2022–present
Electronic Science and Technology · Master's-to-PhD track · Advisor: Huaibin Zheng

Politecnico di Milano · Visiting PhD student — Oct 2026–present
Department of Physics · Supervisor: Antonio Pifferi

Northwest University · B.Eng. — 2018–2022
Electronic Information Science and Technology

Selected Research

The Lift Spectrum · Robust image-free perception — Preprint

arXiv · 2026 · Paper · Code

- Showed that removing reconstruction moves the design problem to the 1D-to-2D lift; organized lifts into a spectrum from fixed-physics inverse to content-adaptive retrieval.
- Proposed STSF + TPLS: +3.2–9.9 pp mIoU over the prior image-free method, near-constant accuracy down to 0.78% sampling, and up to +21.7 pp over clean-trained reconstruct-then-segment at 20 dB noise.

Sparse Hadamard single-pixel imaging

Optics Express · 2025 · Paper · Code

- Gave orthogonal Hadamard patterns tunable sparsity (two acquisition modes, full theory): noise-robust imaging through scattering, low-order matrices that save memory, and no training or post-processing.

LDGI · Optical phased-array imaging

Optics Communications · 2025 · Paper · Code

- Showed that line patterns from one-dimensional optical phased arrays break the noise symmetry assumed by ghost-imaging denoising; proposed LDGI (orders-of-magnitude gains) and validated it optically.

Current Research

Developing acquisition-aware models and adapting pretrained multimodal models to bucket-detector signals. Planned visiting research: time-domain diffuse optics and 3D reconstruction.

Open Source & Tools

Open source · remote-gpu-trainer, an agent skill for Claude Code and Codex that runs, verifies and delivers GPU experiments (60+ GitHub stars)
Tools & lab · PyTorch, MATLAB, COMSOL; DMD/SLM illumination, single-pixel detection, optical alignment and system integration
Device modeling · Electro-optic phased-array geometry modeling and optimization for beam steering

Award

2024 · Second Prize, National Post-Graduate Mathematical Contest in Modeling · Code